

Page 1: Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is the simulation of human intelligence by computers. AI enables machines to perform tasks such as learning, reasoning, problem-solving, and decision-making.

Machine Learning (ML) is a subset of AI that allows computers to learn from data without being explicitly programmed. **Types of Machine Learning**

1. Supervised Learning – Uses labeled data.
2. Unsupervised Learning – Uses unlabeled data.
3. Reinforcement Learning – Learns by rewards and penalties.

Page 2: Deep Learning and Neural Networks

Deep Learning is a subset of machine learning that uses neural networks with multiple layers.

Neural Network consists of an input layer, hidden layers, and an output layer. **Applications of AI**

- Healthcare
 - Finance
 - Education
 - Self-driving cars
 - Chatbots and virtual assistants
- Advantages**

High accuracy, automation, and fast decision-making.

Page 3: Common Terms

Classification predicts categories such as spam or not spam. **Regression** predicts continuous values such as house prices. **Training Data** is the data used to teach a model. **Model** is the learned representation after processing data. **Overfitting** occurs when a model memorizes the training data and performs poorly on new data. **Conclusion**

Machine learning is a powerful technology used in many real-world applications and is an important branch of Artificial Intelligence.